Counterexamples to a circular coloring conjecture for the Cartesian sum

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October 22, 2026

Abstract

Liu and Zhu conjectured that the circular chromatic number of the Cartesian sum, also called the disjunctive or OR product, is at most $\max\{\chi_c(G)\chi(H), \chi(G)\chi_c(H)\}$. We give an infinite family of counterexamples with both factors circular complete graphs. The smallest instance here is $K_{5/2} \oplus K_{11/4}$, a 34-regular graph on 55 vertices with $33/4 < \chi_c \leq 17/2$. The proof reduces a hypothetical coloring to two rigid arrangements of intervals on a circle and excludes both arrangements.

1 Statement

For integers $p \geq 2q > 0$, let $K_{p/q}$ have vertex set $\mathbb{Z}_p$, with distinct x, y adjacent when their cyclic distance is at least q. The Cartesian sum $G \oplus H$ has vertex set $V(G) \times V(H)$; two pairs are adjacent when at least one pair of corresponding coordinates is adjacent. A (P, Q)-coloring is a homomorphism to $K_{P/Q}$. Liu and Zhu [1] conjectured

$$\chi_c(G \oplus H) \leq \max\{\chi_c(G)\chi(H), \chi(G)\chi_c(H)\}. \tag{1}$$

Their theorem concerning the ordinary chromatic number is a distinct statement.[1]

Theorem 1. *Let $q \geq 4$, let $p \in \{3q-2, 3q-1\}$, and assume $\gcd(p, q) = \gcd(q, 3) = 1$. Then*

$$\chi_c\big(K_{5/2} \oplus K_{p/q}\big) > \frac{3p}{q} = \max\{\chi_c(K_{5/2})\chi(K_{p/q}), \chi(K_{5/2})\chi_c(K_{p/q})\}.$$

These graphs have $5p$ vertices and degree $5p - 6q + 3$.

In particular, $p = 3q - 1$ is admissible for every $q \geq 4$ not divisible by 3. This gives infinitely many orders. The $p = 3q - 2$ choice is admissible whenever q is also odd.

Corollary 2. *The graph $X = K_{5/2} \oplus K_{11/4}$ has 55 vertices, 935 edges, and degree 34, and satisfies $33/4 < \chi_c(X) \leq 17/2$ and $\chi(X) = 9$.*

[1] A Lean 4 certificate for the 55-vertex instance is available at the accompanying GitHub repository (commit `a4b5b22`). It verifies the nonexistence of a $(33, 4)$-coloring and the explicit $(17, 2)$-coloring below. The certificate uses Lean and Mathlib version 4.35.0; the obstruction depends only on the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`. The infinite-family theorem and its circular-chromatic interpretation are proved in the text and are outside the formalization's scope.

Proof. Apply Theorem 1 with $(p, q) = (11, 4)$. The coloring $(a, b) \mapsto 3\lfloor a/2 \rfloor + \lfloor b/4 \rfloor$ takes values in $\{0, \ldots, 8\}$ and is proper: each color class is contained in a product of two consecutive blocks, independent in the respective factors. The explicit $(17, 2)$-coloring in Table 1 gives the other circular bound. Since $\chi(X) = \lceil \chi_c(X) \rceil$, the strict lower bound gives $\chi(X) \geq 9$. The degree follows by counting nonneighbors: a vertex has $(2 \cdot 2 - 1)(2 \cdot 4 - 1) = 21$ possible coordinatewise nonneighbors, including itself, out of 55 vertices. Hence its degree is $55 - 21 = 34$. □

$a \backslash b$	0	1	2	3	4	5	6	7	8	9	10
0	0	0	11	11	11	11	6	6	6	0	0
1	16	15	15	10	10	10	5	5	5	5	16
2	15	15	14	14	9	9	9	4	4	4	3
3	2	13	13	13	13	8	8	8	3	3	3
4	1	1	12	12	12	7	7	7	7	1	1

Table 1: A $(17, 2)$-coloring of X: the entry in row a, column b is the color of $(a, b) \in \mathbb{Z}_5 \times \mathbb{Z}_{11}$. Every edge has target cyclic color distance at least 2.

Corollary 3 (identical factors)**.** *The graph* $K_{11/4} \oplus K_{11/4}$ *violates* (1)*. It has* 121 *vertices and* 4356 *edges.*

Proof. The map $a \mapsto \lfloor 11a/5 \rfloor$ sends $K_{5/2}$ homomorphically into $K_{11/4}$. Taking its product with the identity on $K_{11/4}$ sends X into the stated graph, whose circular chromatic number is therefore greater than $33/4$. The proposed right side for these two identical factors is $33/4$. The edge count follows from degree $121 - (2 \cdot 4 - 1)^2 = 72$. □

2 A saturated interval system

We prove the theorem. Put $G = K_{5/2}$, $H = K_{p/q}$, and $P = 3p$. The independent sets of G are singletons and consecutive pairs in a geometric 5-cycle (whose edges are the nonedges of G).

Every independent set of H is contained in a cyclic arc of q consecutive residues. Indeed, translate one of its elements to 0. All other elements lie in $[0, q-1] \cup [p-q+1, p-1]$. If u and $-v$ are the extreme elements on the two sides and $u + v \geq q$, then their compatibility forces $p - (u + v) \leq q - 1$; since $u + v \leq 2q - 2$, this gives $p \leq 3q - 3$, contrary to the hypothesis. Thus $u + v \leq q - 1$. Consequently $\alpha(H) = q$, and its maximum independent sets are precisely its cyclic q-arcs. The identity (p, q)-coloring and the fractional lower bound $|V(H)|/\alpha(H)$ show $\chi_c(H) = p/q$. Three consecutive q-arcs give a 3-coloring, while $p/q > 2$ shows $\chi(H) = 3$. Similarly $\chi_c(G) = 5/2$ and $\chi(G) = 3$. As $p/q \geq 5/2$, the right side of (1) is $3p/q$. Our coprimality assumptions give $\gcd(P, q) = 1$. It suffices to show that there is no (P, q)-coloring of $G \oplus H$: for a finite graph, $\chi_c(F) \leq P/q$ if and only if $F \to K_{P/q}$ [1].

Suppose such a coloring c exists. Replace each color $c(a, b)$ by its interval $I(a, b)$ of q consecutive slots in $\mathbb{Z}_P$. Intervals corresponding to adjacent vertices are disjoint. Write $U_b = \bigcup_{a \in \mathbb{Z}_5} I(a, b)$. We claim $|U_b| \geq 3q$. If U_b is not the whole circle, cut it at an unoccupied slot, and let J be the intersection graph of its five line intervals. Since G is a subgraph of the complement of J, the interval-graph coloring property gives

$$3 = \chi(G) \leq \chi(\overline{J}) = \alpha(J) \leq \left\lfloor \frac{|U_b|}{q} \right\rfloor.$$

The claim is immediate if $U_b = \mathbb{Z}_P$. At a target slot, the active second coordinates form an independent set of H, so at most q fibers are active. Therefore

$$3pq \le \sum_{b\in\mathbb{Z}_p} |U_b| \le Pq = 3pq. \tag{2}$$

Equality gives $|U_b| = 3q < P$ for all b and exactly q active fibers at each slot. They form a cyclic q-arc in H.

Each occupied run of U_b has length divisible by q. For if its run lengths are $\ell_1, \dots, \ell_k$, disjoint length-q intervals fit at most $\sum_i \lfloor \ell_i/q \rfloor$ times, whereas the preceding interval argument requires at least three. Hence

$$3 \le \sum_i \lfloor \ell_i/q \rfloor \le \sum_i \ell_i/q = 3.$$

Partition the runs into q-blocks; there are three for each fiber. Let x_t be the number of blocks beginning at slot $t \in \mathbb{Z}_P$. Every slot is covered by exactly q blocks, giving $\sum_{j=0}^{q-1} x_{t-j} = q$. Subtracting consecutive equations gives $x_t = x_{t-q}$. Since $\gcd(P, q) = 1$, we have $x_t = 1$ for every t. Let b_t denote the second coordinate of the unique block beginning at t. This is a cyclic word of length P; each label occurs three times, and the labels of every q consecutive blocks are distinct and form a cyclic q-arc in H.

3 The two possible block words

Lemma 4. *Up to reversing the cyclic order of H, exactly one of the following forms can arise from the preceding block word:*

1. *Each U_b is the union of three isolated q-runs.*
2. *Each U_b is one $3q$-run, and the p run starts occur every three target slots.*

Proof. At time t consider the ordered queue $(b_t, \dots, b_{t+q-1})$. Its underlying set is a cyclic q-arc. Two successive queues have $q-1$ common labels. Since $p > 2q$, their arcs either agree or differ by shifting one endpoint one place. Let D_t be the net forward displacement of the arc up to time t, lifting its starting label to the integers. Give each queue entry its start-time residue modulo q. The departing and arriving entries have the same residue. If τ_k is the residue initially at spatial position k in the arc, then the residue at position k at time t is τ_{k+D_t}.

After $L = P = 3p$ steps, the ordered queue returns to itself, while all start-time residues have increased by L modulo q. The arc has completed an integer number W of turns, so $D_L = pW$ and

$$\tau_{k+D_L} = \tau_k + L \pmod q. \tag{3}$$

The τ_k are a permutation of $\mathbb{Z}_q$. If $g = \gcd(D_L, q)$, iterating (3) q/g times gives $(q/g)L \equiv 0 \pmod q$; since $\gcd(L, q) = 1$, we must have $g = 1$. Reverse the labeling of H if necessary so that $D_L > 0$, hence $W \ge 1$. Equation (3) also forces $\tau_k = \tau_0 + ak \pmod q$ for some $1 \le a < q$.

Set $z_t = t - \tau_0 - aD_t \pmod q$. A hold advances z by one. A forward move is possible only at $z = 0$ and sends z to $1 - a$; a backward move is possible only at $z = -a$ and sends z to 1. This directed phase graph consists of the ordinary q-cycle of holds and two chords. Its simple directed cycles are: q holds; one forward move and $a - 1$ holds; or one backward move and $q - a - 1$ holds. In particular, if their multiplicities in a closed-walk decomposition are N_0, N_+, N_-, then

$$L = qN_0 + aN_+ + (q-a)N_-, \qquad D_L = N_+ - N_-.$$

Substituting $D_L = pW$ yields

$$3p = apW + q(N_0 + N_-). \tag{4}$$

Thus $1 \le aW \le 3$ and, by $\gcd(p, q) = 1$, $aW \equiv 3 \pmod{q}$. As $q \ge 4$, necessarily $aW = 3$ and $N_0 = N_- = 0$. Either $(a, W) = (1, 3)$, a forward step every slot, or $(a, W) = (3, 1)$, a forward step every three slots. In the first case each label traverses the arc three times, occupying three isolated q-runs. In the second it traverses the arc once, occupying a single $3q$-run, with new runs starting every three slots. □

4 Excluding both forms

Lemma 5 (cyclic interval expansion)**.** *Let A be a nonempty subset of $\mathbb{Z}_L$, and let $1 \le r \le L$. The union of the length-r cyclic intervals starting at the elements of A has size at least*

$$\min\{L, |A| + r - 1\}.$$

Proof. Set $B_0 = A$ and $B_{j+1} = B_j \cup (B_j + 1)$. A nonempty proper subset of $\mathbb{Z}_L$ cannot be invariant under translation by one. Consequently each step increases the cardinality by at least one until the whole circle is occupied. The required union is B_{r-1}. □

Consider the first form of Lemma 4. Each vertex interval must coincide with one of its fiber's three isolated q-blocks. Fix $a \in \mathbb{Z}_5$. The p vertices (a, b), with $b \in \mathbb{Z}_p$, have distinct interval starts: a slot is the start of exactly one block, and that block belongs to exactly one fiber. Thus, writing

$$V_a = \bigcup_{b \in \mathbb{Z}_p} I(a, b),$$

Lemma 5 gives $|V_a| \ge p + q - 1$, since $p + q - 1 < P = 3p$.

At any target slot, the active first coordinates form an independent set of $K_{5/2}$, so there are at most two. Double counting gives

$$5(p + q - 1) \le \sum_{a \in \mathbb{Z}_5} |V_a| \le 2P = 6p.$$

It follows that $5(q - 1) \le p$. This contradicts $p \le 3q - 1 < 5(q - 1)$ for $q \ge 4$, excluding the first form.

Lemma 6 (three shifted segments)**.** *Suppose three proper assignments of length-q intervals to the vertices of $K_{5/2}$ are contained in respective segments of length $3q$, starting at $0, d, 2d$, where $0 < d < q$. If intervals associated with adjacent vertices in different assignments are also disjoint, then $2d = q$.*

Proof. For any one assignment, orient each edge of $K_{5/2}$ toward the larger interval start. An odd cycle has a directed two-edge path. Its two consecutive start differences are at least q, while all starts lie in an interval of length $2q$. The three starts must therefore be $0, q, 2q$. Label those vertices $0, 2, 4$ in the complementary 5-cycle. The other starts have the form $x, q + y$, where $0 \le x \le y \le q$.

Normalize the first assignment to have anchors $0, 2, 4$ at $0, q, 2q$. The first anchor of the second assignment, at d, overlaps the first anchors 0 and 2, so its label must be 1, the unique label nonadjacent to both. Its middle anchor at $q + d$ overlaps anchors 2 and 4 of the first assignment and must have label 3. Repeating this reasoning, the third assignment's first anchor at $2d$ has label 2. It must be disjoint from the first assignment's anchors 0 at 0 and 4 at $2q$. These two requirements give $2d \ge q$ and $2d \le q$. □

In the second form of Lemma 4, three successive fiber runs start at $0, 3, 6$ after translation. Their $3q$-segments fit in a line because $3q + 6 < 3p$. Intervals for adjacent first coordinates in different fibers are disjoint by the definition of the Cartesian sum. Lemma 6, with $d = 3$, forces $q = 6$, contradicting $\gcd(q, 3) = 1$. Finally, each vertex has $3(2q - 1)$ coordinatewise nonneighbors, including itself, among the $5p$ vertices; hence its degree is $5p - 6q + 3$. This finishes the proof of Theorem 1. □

References

[1] D. D.-F. Liu and X. Zhu, Coloring the Cartesian sum of graphs, *Discrete Mathematics* **308** (2008), 5928–5936. doi:10.1016/j.disc.2007.11.005.